

Controlling DRONES And UAVs: Advancements In WIRELESS TECHNOLOGIES

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Upgraded wireless communication systems, latest navigation technologies and advanced flight control programs are the key ingredients for advanced applications of drones and UAVs

Wireless technology plays a major role in controlling unmanned aerial vehicles (UAVs) and drones, by offering safe flying operations. Mesh radio is increasingly being employed for commercial and military use. It enables drones to communicate with other drones, vehicles and human beings. Stable command and control link with the pilot on ground are needed for safely operating drones.

Range extender technology is used to extend the range of communication between a smartphone or tablet and the drones in an open, unobstructed area. Transmission distances can reach up to 700 metres. Range extenders have unique media access control (MAC) addresses and network names (SSIDs).

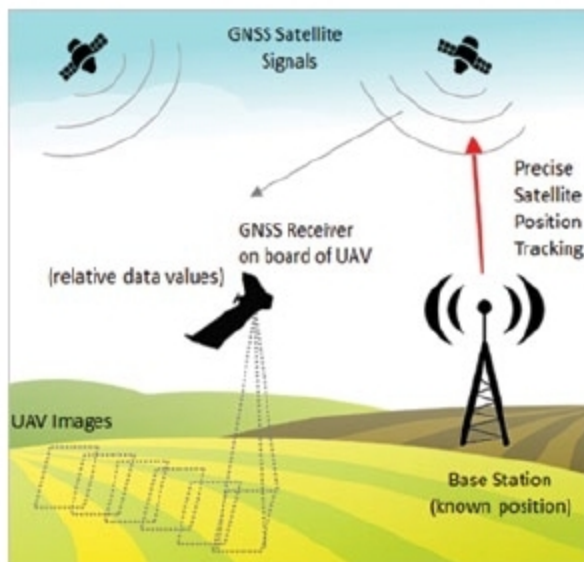


Fig. 1: GNSS for drones
(Credit: www.questuav.com)

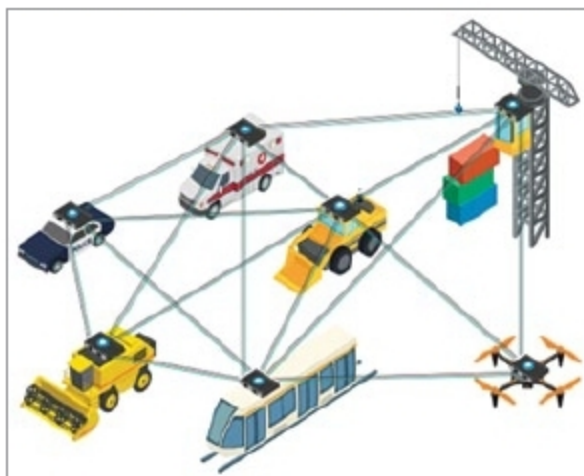


Fig. 2: Breadcrumbs communicating peer-to-peer
(Credit: www.rajant.com)

Navigation technologies for wireless control

Drones use highly-accurate navigation systems for surveying landscapes, search-and-rescue operations and building 3D maps, among others. These can fly in both satellite and non-satellite modes. Drones utilise Global Navigation Satellite System (GNSS) to track their position and navigate in self mode.

GNSS is a satellite system that is used to pinpoint the geographical location anywhere through a receiver. It utilises a constellation of orbiting satellites working in conjunction, through ground station networks. Each satellite transmits coded signals at fixed intervals. This signal information is converted by the receiver into position, time and velocity, which is later utilised to calculate the exact position of the transmitting satellite and its distance.

A drone, which is ready to fly, records the home coordinates once it receives enough signal

from GNSS satellites. Some drones seek location through GPS satellites. This allows drones to fly in ready-to-fly mode when their compasses are calibrated and these find more than six satellites.

Latest wireless communication technologies

Vikas Jain, team leader, Cyient, says, "Latest wireless technology used is Wi-Fi in 2.4GHz to 5.47GHz frequency band, for general control and general sensor-like drone payloads. Where payloads like video and cinematography are concerned, latest technologies include first person view (FPV), ocusync and light bridge video transmission. These technologies meet cutting-edge demand for long distances, and provide very high-quality video transmission in HD.

"Developments and possibilities of using 7GHz frequency band and a dedicated frequency band for drone application are also under development. LoRa-based drone technology development can provide good meshing and networking options for drones."

Don Gilbreath, Rajant Corp., says "We provide rugged mesh radios for military and extreme temperature commercial locations like open pit mines, oil fields and ship ports.

"With regards to UAVs, we have DX-1 development kit. It is a small, light radio with antennae that weigh 600 grams. We provide extremely strong encryption options that offer different frequencies—licensed, unlicensed and military bands—which help with high performance use. We test detection technologies for video, lidar, radar and sonar, too. Real-time kinematics for UAVs is compatible with existing customers' survey-grade base stations. It yields two-to-three-centimetre location accuracy, which is much better than GPS native."

New wireless connectivity options

Kinetic mesh (KM) enables industrial sectors (mining, military, oil and gas, and public safety) to establish highly-reliable and